What the powers of an adjacency matrix count, and how the matrix shows that a graph is connected

A study note for the graph theory course

November 5, 2026

Abstract

A graph with N vertices can be written as an N by N table of zeros and ones, its adjacency matrix A. This note answers three questions about that table. First, the entry in row i and column j of the k-th power of A counts the walks of length k from vertex n_i to vertex n_j. It counts walks, not paths. Second, a graph with $N \geq 2$ vertices is connected exactly when every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, and equivalently when $(I + A)^{N-1}$ has no zero entry. Third, under the course convention a path between two vertices is a walk from one to the other with no repeated vertex. Each result is proved, shown on a connected and on a disconnected graph with three vertices, and compared with the textbook's definitions and numbering. The note is written for a first reading, and each definition and theorem is introduced by a short paragraph in plain words.

1 Introduction

1.1 The three questions and where they came from

A lecture board shows a small drawing with three points labelled 1, 2 and 3, and next to it a table of zeros and ones with the rows 0 1 0, 1 0 1 and 0 1 0 [2, 3]. The board says that a graph with N nodes is given through this table, called its adjacency matrix and written A, and then asks two questions [2]. A third question comes from the assignment text, which the user reported to us.

Q1 (board, question 1) What does the entry in row i and column j of a power of A represent? The board writes a small mark above the letter A; we read it as a power k (convention 2).

Q2 (board, question 2) How can one tell from A whether the graph is connected?

Q3 (assignment text, as reported) What is the definition of a path between two vertices n_i and n_j? The board uses the word *path* in its definition of connected but does not define it [2].

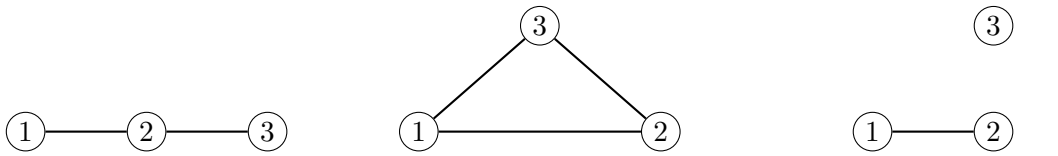
Short answers, with every word defined in Section 2. Q1: the entry counts the walks of exactly k roads from n_i to n_j (Theorem 4.4); it counts walks, not paths (Proposition 4.7). Q2: the graph is connected exactly when a sum of powers of A has no zero off the diagonal (Theorem 5.5). Q3: a path is a walk that never visits a vertex twice (Definition 2.15). The order of the note follows what each answer needs. The words come first (Section 2), then the comparison of path conventions (Section 3), then Q1 (Section 4) and Q2 (Section 5), and finally the limits (Section 6) and the conclusion (Section 7).

1.2 The story used in the plain-words paragraphs

Before every definition, theorem and worked example there is a short story paragraph, marked in bold at its start. The story is this. A country has towns and roads. Each road joins two different towns. Two towns share at most one road. People make trips along the roads. Figure 1 shows the country of the board. It has three towns in a row. A road joins towns 1 and 2, and a road joins towns 2 and 3. The story paragraphs use only the story words of table 1.

Table 1: The story dictionary: one story word for each technical term.

Technical term	Story word
graph	map
vertex	town
edge	road
adjacent vertices	neighbour towns
adjacency matrix	road table
walk	trip
length of a walk	number of roads on the trip
trail	trip that uses no road twice
path	trip that visits no town twice
cycle	round trip that uses no road twice
degree of a vertex	number of roads at a town
distance	fewest roads needed between two towns
connected graph	every town can be reached from every other
component	island of towns



(a) board map (b) drawing read as three roads (c) one town alone

Figure 1: The three small maps used in this note. Map (a) is the one given by the matrix on the board. Map (b) is the drawing on the board read literally. Map (c) is a map that is not connected.

1.3 Conventions and assumptions

The board and the textbook differ in places, and the board has one mark that cannot be read with certainty. We fix the following conventions. They are cited later by their number.

1. *Finite simple undirected graphs.* Every graph is finite, has no road from a town to itself, no two roads between the same two towns, and no one-way roads. The user stated this setting, and the symmetric, zero-diagonal matrix on the board fits it [2].

2. *The mark above A is a power k.* The mark is a small letter between quotation-like strokes that could be read as k or n [2, 3]. The user's text asks about powers A^k, so we read k. This is an assumption, not something read off the board.

3. *The matrix, not the drawing, is the example.* The drawing on the board shows three lines: 1–2 (with a stray arrowhead at 2), 1–3 and 2–3. The matrix has a 1 only for the pairs $1, 2$ and $2, 3$ [2, 3]. The two disagree on the pair $1, 3$. The problem statement says the graph is given through its matrix, so fig. 1(a) is the example, and the arrowhead is read as a stray mark because the matrix is symmetric. The drawing's three-road map (fig. 1(b)) is also computed, as a side example.

4. *Notation.* The board writes (G, V) for a graph and $e_{ij} \in V$ for "there is a road between n_i and n_j". The textbook writes $G = (V, E)$ with V the set of vertices and E the set of edges, so the board's V in that sentence is the textbook's E [2, 7]. We write $G = (V, E)$, $V = \{n_1, \ldots, n_N\}$, $N = |V|$, and e_{ij} or $n_i n_j$ for the edge joining n_i and n_j. The textbook calls the vertices v_i; this note uses n_i as on the board. The textbook's page numbers in this note are printed page numbers, which are one less than the page numbers of the PDF file.

5. *The word "path".* Under the course convention, stated by the user, a path has no repeated vertex. This is the textbook's *simple path*. The textbook's own word "path" means something weaker. Table 2 compares the two.

6. *Walks of length 0.* We allow a walk that is one vertex and no edge. The textbook's definition does not say whether this is allowed [7]; the choice is ours, made so that A^0, the identity matrix, also counts walks.

7. *The five-town map.* Several examples use the map of fig. 2. It is drawn after the textbook's Fig. 1.2.5 [7, p. 21]. The textbook prints no list of edges, so we read the edge list from the page image. All the textbook's statements about that figure that we checked hold on this reading.

The notation used throughout is as follows. $A = (A_{ij})$ is the adjacency matrix; A^k is its k-th power and $(A^k)_{ij}$ the entry in row i and column j of that power; I is the identity matrix; $d(n_i)$ is the degree of n_i; $S = A + A^2 + \cdots + A^{N-1}$ is the sum used in the test of Section 5. The letters N (number of vertices), k (length of a walk or exponent), i, j, t (indices) and S keep one meaning throughout.

1.4 What the sources contain

The textbook [7] defines the adjacency matrix, walks, trails, paths, cycles, distance and connectedness, and we use its numbers. A search of the text layer of the whole book for the word "matrix" finds the adjacency matrix only on printed pages 7, 11, 12, 34 and 61, and finds no theorem about the powers of A and no matrix test for connectivity. The text layer drops most symbols, so this shows that we did not find such a statement, not that none exists. We therefore prove both results in this note. Grinberg's lecture notes [1] state the walk-counting theorem for multidigraphs, which are more general than simple graphs, and give the walk-to-path lemma with a length bound. Paramadevan and Sotheeswaran [5] state the connectivity test as their Theorem 2. We use their paper for the statement only; their Theorem 1 and the proof of their Theorem 2 are discussed in Proposition 4.7 and Remark 5.6. An encyclopedia page [6] is used only to name the English words for the textbook's two kinds of path. What each source supports is listed in Appendix B.

2 Background: graphs, walks and paths

Throughout this section $G = (V, E)$ is a graph as in convention 1, $V = \{n_1, \ldots, n_N\}$, and u, v, w denote vertices. The definitions follow the textbook's order [7]: graph, adjacent, adjacency matrix, degree, walk, closed walk, trail, path, cycle, simple cycle, distance. After each definition come a remark on what it allows and forbids, an example and a non-example. Figure 2 is the map for the examples of walks, trails, paths and cycles.

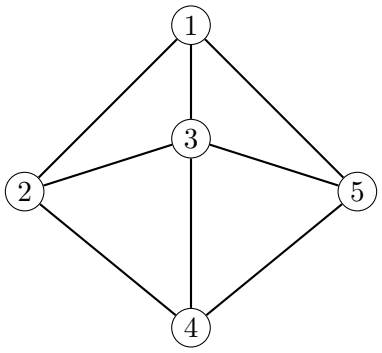


Figure 2: The five-town map used for the examples of this section, drawn after the textbook's Fig. 1.2.5 (convention 7). The label i stands for the vertex n_i; these are the textbook's $v_1, \ldots, v_5$. The roads are n_1n_2, n_1n_3, n_1n_5, n_2n_3, n_2n_4, n_3n_4, n_3n_5 and n_4n_5.

2.1 Graphs, adjacency, the adjacency matrix, degree

In plain words. A map is a set of towns and a set of roads. Each road joins two different towns. No two roads join the same pair of towns. A road can be driven in both directions.

Definition 2.1 (simple graph)**.** A *simple undirected graph* $G = (V, E)$ consists of a finite set V of *vertices* and a set E of *edges*, each edge being a set $\{n_i, n_j\}$ of two different vertices, written n_in_j or e_{ij}. Hence $n_in_j = n_jn_i$. The number of vertices is $N = |V|$. The textbook uses the notation $G = (V, E)$ [7].

Remark 2.2. Because an edge is a set of two different vertices, no edge joins a vertex to itself (no *loop*), and because E is a set, no two edges have the same two ends (no *multiple edges*). Edges have no direction.

Example. The board map of fig. 1(a): $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2, n_2n_3\}$, so $N = 3$.
Non-example. A road from n_1 to n_1, or two roads between n_1 and n_2, would not be allowed.

In plain words. Two towns are neighbours when one road joins them directly.

Definition 2.3 (adjacent; textbook Definition 1.1.4)**.** The vertices n_i and n_j are *adjacent* if $n_in_j \in E$ [7, Definition 1.1.4, p. 11].

Remark 2.4. Adjacency is a statement about one edge only. It is symmetric, and in a simple graph no vertex is adjacent to itself.

Example. In fig. 1(a), n_1 and n_2 are adjacent.
Non-example. In fig. 1(a), n_1 and n_3 are not adjacent: the road between them exists only in the drawing of fig. 1(b) (convention 3).

In plain words. The road table has one row and one column for each town. A one says the two towns are neighbours. A zero says they are not.

Definition 2.5 (adjacency matrix)**.** The *adjacency matrix* of G is the $N \times N$ matrix $A = (A_{ij})$ with $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise. The textbook defines $A(G)$ in the same way, without a number of its own, after its Definition 1.1.6 [7, p. 11], and the board writes it as A_{ij} [2].

Remark 2.6. The matrix records only whether a road exists, never how many; for a simple graph nothing is lost. The textbook notes that $A_{ii} = 0$ for every i and $A_{ij} = A_{ji}$ for all i, j [7, p. 11]: the diagonal is zero because there are no loops, and the matrix is symmetric because edges have no direction.

Example. The board matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

is the adjacency matrix of fig. 1(a): $A_{12} = A_{21} = 1$ and $A_{23} = A_{32} = 1$, all other entries 0.
Non-example. The matrix with rows 1 1 0, 1 0 0, 0 0 0 is not the adjacency matrix of a simple graph, because $A_{11} = 1$ would be a loop. A matrix with $A_{12} = 1$ and $A_{21} = 0$ is not the adjacency matrix of an undirected graph, because it is not symmetric.

In plain words. The number of roads at a town tells how busy the town is.

Definition 2.7 (degree; textbook Definition 1.2.1)**.** The *degree* $d(n_i)$ of a vertex n_i is the number of edges at n_i [7, Definition 1.2.1, p. 16].

Remark 2.8. The textbook introduces degree after the adjacency matrix, and this note keeps that order. In a simple graph the edges at n_i correspond one to one to the vertices adjacent to n_i, so the degree is also the number of neighbours.

Example. In fig. 1(a), $d(n_1) = 1$, $d(n_2) = 2$ and $d(n_3) = 1$.
Non-example. The number $N = 3$ of vertices is not a degree: it counts towns, not roads at one town.

2.2 Walks, trails, paths and cycles

In plain words. A trip goes from a town along a road to the next town, and so on. The trip may come back to a town it already visited. It may use a road again. Count the roads a trip uses. A road used twice is counted twice.

Definition 2.9 (walk; textbook Definition 1.2.10)**.** A *walk of length* k from u_0 to u_k is a sequence $u_0, e_1, u_1, \ldots, u_{k-1}, e_k, u_k$ of vertices and edges with $e_t = u_{t-1}u_t$ for $1 \leq t \leq k$ [7, Definition 1.2.10, p. 21]. Its *length* is k.

Remark 2.10. Vertices and edges may repeat in a walk, and the length counts every edge each time it occurs [7, p. 22]. In a simple graph the edge between two adjacent vertices is the only one, so a walk is determined by its vertices. The textbook also writes a walk as the sequence of its vertices [7, p. 21], and so does this note: a walk of length k is a list $u_0, \ldots, u_k$ of $k + 1$ vertices in which each vertex is adjacent to the one before it. By convention 6 a single vertex is a walk of length 0.

Example. On the five-town map (fig. 2), $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is a walk of length 8. It repeats vertices and edges.
Non-example. On the same map, $n_3, n_1, n_4, n_2, n_3, n_5$ is not a walk, because n_1n_4 is not an edge.

In plain words. A round trip ends in the town where it began.

Definition 2.11 (closed walk; textbook Definition 1.2.11)**.** A walk $u_0, \ldots, u_k$ is *closed* if $u_0 = u_k$ [7, Definition 1.2.11, p. 22].

Remark 2.12. Nothing else is required: a closed walk may repeat vertices and edges.

Example. On the board map (fig. 1(a)), n_1, n_2, n_1 is a closed walk of length 2.
Non-example. On the same map, n_1, n_2, n_3 is a walk but not a closed one.

In plain words. Some trips use no road twice. A town may still be visited twice.

Definition 2.13 (trail; textbook Definition 1.2.12)**.** A *trail* is a walk whose edges $e_1, \ldots, e_k$ are pairwise distinct. The textbook calls this a "path" [7, Definition 1.2.12, p. 22]; "trail" is the English word [6, 1].

Remark 2.14. A trail may visit a vertex more than once, but it may not use an edge more than once. To avoid a clash of words, this note writes *trail* for the textbook's "path" of Definition 1.2.12 each time.

Example. On the five-town map, $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail. It visits n_3 twice but uses each of its five roads once.
Non-example. The walk $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is not a trail: it uses the roads n_2n_3, n_2n_4 and n_3n_4 twice each.

In plain words. A trip that visits no town twice is the most careful kind of trip.

Definition 2.15 (path between n_i and n_j; textbook Definition 1.2.13, "simple path")**.** Let $n_i, n_j \in V$. A *path between n_i and n_j* is a walk $w_0, w_1, \ldots, w_k$ with $w_0 = n_i$, $w_k = n_j$ and $w_0, \ldots, w_k$ pairwise distinct. The number k is its *length*.

This is the answer to question Q3. For $n_i \neq n_j$ it is the textbook's *simple (n_i, n_j)-path*, Definition 1.2.13 [7, p. 22], which the textbook states as a trail whose vertices are pairwise distinct; Proposition 3.1 shows that the condition on the edges is then automatic. It is also the English word "path" [6, 1]. The user stated that this is the sense of "path" in the course (convention 5).

Remark 2.16. A path never repeats a vertex, so it never repeats an edge either (Proposition 3.1). A path between a vertex and itself has only one vertex, and so has length 0: by convention 6 it is the sequence n_i alone. A path of length k has $k + 1$ different vertices.

Example. On the five-town map, n_1, n_3, n_4, n_5 is a path between n_1 and n_5 of length 3. Another path between the same towns is n_1, n_5, of length 1.
Non-example. $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail but not a path, because n_3 repeats.

In plain words. A round trip ends in the town where it began. Here we look at round trips that use no road twice. Such a trip may still pass through a town more than once.

Definition 2.17 (cycle; textbook Definition 1.2.14)**.** A *cycle* is a closed trail, which the textbook words as a closed "path" [7, Definition 1.2.14, p. 22].

Remark 2.18. A cycle may visit a vertex more than once, but it may not use an edge more than once. The word "cycle" is used here in the textbook's sense, which is broader than a simple cycle (Definition 2.19).

Example. On the five-town map, $n_3, n_2, n_1, n_3, n_5, n_4, n_3$ is a cycle.
Non-example. n_1, n_3, n_2 is a trail that is not closed, so it is not a cycle.

In plain words. Some round trips pass through no town twice before they end. The textbook calls them simple.

Definition 2.19 (simple cycle; textbook Definition 1.2.15)**.** A cycle is *simple* if only its first and last vertices are equal [7, Definition 1.2.15, p. 22].

Remark 2.20. In a simple graph a simple cycle has length at least 3. A cycle of length 1 would need a loop. A cycle of length 2 would be u, v, u, which uses the edge uv twice, against the definition of a trail. The textbook writes C_n for the simple cycle on n vertices, with $n \geq 3$ [7, p. 22]. To prevent confusion this note writes "simple cycle" whenever that is meant.

Example. On the five-town map, n_1, n_3, n_4, n_2, n_1 is a simple cycle of length 4.
Non-example. $n_3, n_2, n_1, n_3, n_5, n_4, n_3$ is a cycle but not a simple cycle, because n_3 repeats inside it.

In plain words. Pick two towns. Look at all trips between them that visit no town twice. The fewest roads on such a trip is the answer. If there is no such trip, we say the answer is endless.

Definition 2.21 (distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ between u and v is the length of a shortest path from u to v if there is one, and $+\infty$ otherwise.

Remark 2.22. The textbook states the definition with its own word "path", that is, with trails [7, Definition 1.2.16, pp. 22–23]. Proposition 3.4 shows that the value is the same with paths, with trails and with walks. A distance is a number or $+\infty$, never a negative number.

Example. On the board map, $d(n_1, n_3) = 2$: the only path between them is n_1, n_2, n_3.
Non-example. On the map of fig. 1(c), $d(n_1, n_3) = +\infty$, which is not a number of roads.

3 Question 3: the path between two vertices

The answer to Q3 is Definition 2.15, in its place in the textbook's order: a path between n_i and n_j is a walk from n_i to n_j with no repeated vertex. This section adds only what is new: the comparison with the other meanings of the word, and three facts that the later sections use. The facts are that every path is a trail and every trail is a walk (Proposition 3.1), that every walk contains a path with the same ends and no greater length (Lemma 3.2), and that a path has at most $N - 1$ edges (Proposition 3.4).

3.1 Three meanings of one word

The textbook uses the word "path" for a walk with no repeated edge, and needs the extra word "simple" for a walk with no repeated vertex [7, Definitions 1.2.12 and 1.2.13, p. 22]. English texts call the first a trail and the second a path [6, 1]. The course meaning of "path" is the second one (convention 5), which is also the English one. A statement quoted from any source must be translated through table 2 before it is used.

Table 2: The three conventions. The last column is the word used in this note.

What the trip may not do	Textbook word	Textbook number	English word	This note
nothing is forbidden	walk	1.2.10	walk	walk
repeat an edge	"path"	1.2.12	trail	trail
repeat a vertex	"simple path"	1.2.13	path	path

3.2 Path, trail, walk

In plain words. Every trip that visits no town twice also uses no road twice. A trip that uses no road twice is still a trip. The reverse does not hold. A trip can visit a town twice and use each road once.

Proposition 3.1. *Every path is a trail and every trail is a walk. Neither statement can be reversed.*

Proof. A trail is a walk by Definition 2.13. Let $u_0, \ldots, u_k$ be a path, so its vertices are pairwise distinct, and suppose two of its edges are equal: $e_a = e_b$ with $a < b$. Then $\{u_{a-1}, u_a\} = \{u_{b-1}, u_b\}$, so u_{a-1} is equal to u_{b-1} or to u_b. Both of these have an index larger than $a - 1$, because $b - 1 \geq a$. That contradicts the distinctness of the vertices. So the edges are distinct and the path is a trail. For the reverse, on the five-town map (fig. 2), $n_1, n_3, n_2, n_4, n_3, n_5$ is a trail that is not a path, and $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ is a walk that is not a trail (the examples of Definition 2.13 and Definition 2.15). The textbook gives the same two examples [7, p. 22]. □

3.3 A walk contains a path

In plain words. Take any trip from one town to another. If it passes a town twice, cut out the loop between the two visits. The trip is now shorter. It still starts and ends where it did. Keep cutting. In the end no town is visited twice. So a trip between two towns can always be cut down. The cut trip visits no town twice. It is never longer than the first.

Lemma 3.2 (a walk contains a path)**.** *Let $u_0, \ldots, u_k$ be a walk from u to v, with $u = v$ allowed and $k \geq 0$. Then there is a path from u to v of length at most k, whose vertices are some of $u_0, \ldots, u_k$ in the same order.*

The textbook proves this for $u \neq v$ without a bound on the length (Lemma 1.2.1, [7, p. 23]); Grinberg proves the form with the bound, for $u = v$ as well [1, Corollary 2.9.7, p. 39]. The proof below is ours.

Proof. By strong induction on k.

Base case, $k = 0$. The walk is the single vertex $u = v$, which is a path of length 0.

Induction hypothesis. For some $k \geq 1$, the claim holds for every walk of length less than k.

Step. Let $u_0, \ldots, u_k$ be a walk. If its vertices are pairwise distinct, it is a path of length k, and we are done. Otherwise $u_a = u_b$ for some $a < b$. Delete $u_{a+1}, \ldots, u_b$. The result is $u_0, \ldots, u_a, u_{b+1}, \ldots, u_k$ (if $b = k$ it ends at $u_a = u_k = v$). It is still a walk: the only new pair of neighbours is u_a, u_{b+1}, and since $u_a = u_b$ and u_b is adjacent to u_{b+1}, they are adjacent. It has the same ends, it uses vertices of the original walk in the same order, and its length is $k - (b - a) < k$. By the induction hypothesis it contains a path from u to v of length at most $k - (b - a) \leq k$. □

Example: on the five-town map the walk $n_1, n_3, n_2, n_4, n_3, n_2, n_4, n_3, n_5$ of length 8 contains the path n_1, n_3, n_5 of length 2. On the board map the closed walk n_1, n_2, n_1 contains the path n_1 of length 0.

In plain words. Suppose you can reach a town by some trip. Then you can reach it by a trip that visits no town twice. The same holds for a trip that uses no road twice. So the kind of trip does not matter. We only ask whether the town can be reached.

Corollary 3.3 (the three readings agree)**.** *For vertices u and v the following are equivalent: (a) there is a walk from u to v; (b) there is a trail from u to v; (c) there is a path from u to v.*

Proof. (c) implies (b) and (b) implies (a) by Proposition 3.1. (a) implies (c) by Lemma 3.2. □

Grinberg states the equivalence of (a) and (c) [1, Proposition 2.9.10, p. 42]; the step through trails, which the textbook's definition of connected needs, is ours. The consequence is that the textbook's definition of connected, which uses its "path", the board's definition, and the reading with paths without repeated vertices describe the same graphs (Section 5). Only the vocabulary changes.

3.4 Distance and the longest path

In plain words. The fewest roads between two towns is the same whichever kind of trip we use. A trip that visits no town twice can use at most one road less than there are towns.

Proposition 3.4. *Let u, v be vertices of a graph with N vertices.*

(a) If there is a walk from u to v, then the shortest walk, the shortest trail and the shortest path from u to v have the same length.

(b) Hence $d(u, v) = +\infty$ exactly when there is no walk from u to v.

(c) Every path has length at most $N - 1$. Hence a finite $d(u, v)$ is at most $N - 1$.

Proof. (a) Let k be the length of a shortest walk. By Lemma 3.2 there is a path of length at most k, so of length exactly k by minimality. A path is a trail and a trail is a walk (Proposition 3.1), so the shortest path, trail and walk lengths are all squeezed to k. (b) follows from (a) and Corollary 3.3. (c) A path of length k has $k + 1$ pairwise distinct vertices, and G has only N vertices, so $k + 1 \leq N$ [1, Section 2.9.4, p. 40]. □

Remark 3.5. By (a), the textbook's distance, defined with its "path" (a trail), has the same value as Definition 2.21.

In plain words. Here is the board country, with three towns in a row.

Example 3.6 (paths on the board map)**.** On the board map (fig. 1(a)) the only path from n_1 to n_3 is n_1, n_2, n_3, of length 2, and the only path from n_1 to itself is n_1, of length 0. The sequence n_1, n_2, n_1 is a walk of length 2 from n_1 to n_1 but not a path. The distances are $d(n_1, n_2) = 1$, $d(n_1, n_3) = 2$ and $d(n_2, n_3) = 1$. On the map of fig. 1(c) there is no walk from n_3 to n_1 or to n_2, so $d(n_1, n_3) = d(n_2, n_3) = +\infty$ by Proposition 3.4(b).

4 Question 1: what the entries of a power of A count

4.1 Matrix powers

In plain words. To combine the road table with itself, pick a middle town. Ask whether the first town is a neighbour of the middle town. Ask whether the middle town is a neighbour of the last. Do this for every possible middle town and add up the answers.

Definition 4.1 (matrix product and power)**.** For $N \times N$ matrices X and Y, the product XY is the $N \times N$ matrix with

$$(XY)_{ij} = \sum_{t=1}^{N} X_{it} Y_{tj}. \tag{1}$$

The identity matrix I has $I_{ii} = 1$ and $I_{ij} = 0$ for $i \neq j$, and $XI = IX = X$. The powers of A are $A^0 = I$ and $A^{k+1} = A^k A$ for $k \geq 0$.

Remark 4.2. Each entry of a product is a sum over a middle index t. Since A has only the entries 0 and 1, every entry of every power of A is a non-negative integer. The exponent k counts how many copies of A are multiplied; $A^0 = I$ is the case of no copies.

Example. For the board matrix, $(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1 = 2$.
Non-example. $(A^2)_{22}$ is not $A_{22} \cdot A_{22} = 0$. Squaring the matrix is not squaring each entry.

4.2 Basic properties of A

In plain words. The road table has a few simple habits. A town is never its own neighbour. If one town is a neighbour of another, the second is a neighbour of the first. Adding along a row tells how many roads leave that town.

Proposition 4.3 (properties of A and its powers)**.** *Let A be the adjacency matrix of a simple graph.*

(1) $A_{ii} = 0$ *and* $A_{ij} = A_{ji}$ *for all* i, j *[7, p. 11].*

(2) Every power of A is symmetric: $(A^k)_{ij} = (A^k)_{ji}$.

(3) The i-th row sum of A is the degree: $\sum_j A_{ij} = d(n_i)$*. The column sums are equal to the row sums.*

(4) $(A^2)_{ii} = d(n_i)$.

Proof. (1) is the textbook's remark after the definition of $A(G)$, restated in Definition 2.5. (2) Using $(XY)^T = Y^T X^T$ repeatedly and $A^T = A$, we get $(A^k)^T = (A^T)^k = A^k$. (3) The sum $\sum_j A_{ij}$ counts the vertices adjacent to n_i. In a simple graph these correspond one to one to the edges at n_i, so the sum is $d(n_i)$ by Definition 2.7; by (1) the column sums are the same. (4) By (1) and (1), $(A^2)_{ii} = \sum_t A_{it}A_{ti} = \sum_t A_{it}^2 = \sum_t A_{it} = d(n_i)$, because $A_{it} \in \{0, 1\}$ gives $A_{it}^2 = A_{it}$. □

Paramadevan and Sotheeswaran state (4) in the proof of their Theorem 1 [5, p. 17]. The textbook states (1) and does not state (2) to (4), so those are ours. On the board matrix the row sums are $1, 2, 1$, which are the degrees of Definition 2.7's example.

4.3 The walk-counting theorem

In plain words. Multiply the road table by itself a few times. Each number in the result tells how many trips of exactly that many roads lead from one town to another. Trips may come back to a town they visited. Trips may use a road again. Every such trip is counted.

Theorem 4.4 (walk counting)**.** *Let G be a simple undirected graph with vertices $n_1, \ldots, n_N$ and adjacency matrix A. For every integer $k \geq 0$ and all $i, j \in \{1, \ldots, N\}$, the entry $(A^k)_{ij}$ equals the number of walks of length k from n_i to n_j.*

Grinberg proves the corresponding theorem for multidigraphs, which includes $k = 0$ [1, Theorem 4.5.10, pp. 120–123]. The textbook has no such theorem (Section 1.4). The proof below is ours and is written for simple graphs.

Proof. Write $W_k(i, j)$ for the number of walks of length k from n_i to n_j. By Remark 2.10, $W_k(i, j)$ is the number of sequences $u_0, \ldots, u_k$ of vertices with $u_0 = n_i$, $u_k = n_j$ and u_{s-1} adjacent to u_s for $1 \leq s \leq k$. We prove $W_k(i, j) = (A^k)_{ij}$ by induction on k.

Base case, $k = 0$. The only walk of length 0 that starts at n_i is the single vertex n_i. So $W_0(i, j) = 1$ if $i = j$ and 0 otherwise, which is $I_{ij} = (A^0)_{ij}$. (For $k = 1$ directly: a walk of length 1 from n_i to n_j is one edge, so $W_1(i, j) = A_{ij}$.)

Induction hypothesis. For some $k \geq 0$, $W_k(i, t) = (A^k)_{it}$ for all i, t.

Step. Take a walk $u_0, \ldots, u_{k+1}$ of length $k + 1$ from n_i to n_j, and let $n_t = u_k$ be its next-to-last vertex. Dropping the last vertex leaves a walk of length k from n_i to n_t, and n_t is adjacent to n_j. Conversely, a walk of length k from n_i to n_t, followed by n_j when n_t is adjacent to n_j, is a walk of length $k + 1$ from n_i to n_j. Different t give different walks, because t can be read off the walk. Hence

$$W_{k+1}(i, j) = \sum_{t=1}^{N} W_k(i, t)\, A_{tj},$$

because $A_{tj} = 1$ allows exactly one last step from n_t to n_j and $A_{tj} = 0$ allows none. By the induction hypothesis the right side is $\sum_t (A^k)_{it} A_{tj} = (A^{k+1})_{ij}$ by (1). This closes the induction. □

Remark 4.5. The theorem counts walks of exactly length k, and it needs no assumption that the graph is connected. A zero entry $(A^k)_{ij} = 0$ says only that no walk of length exactly k joins the two vertices; a walk of another length may exist. For $k = 2$ and $i = j$ the theorem restates Proposition 4.3(4): $(A^2)_{ii}$ counts the closed walks n_i, n_t, n_i, one for each neighbour n_t.

4.4 Worked example on the board matrix

In plain words. Here is the board country with three towns in a row. We multiply the road table by itself and read a few numbers. Each number counts trips of the chosen number of roads.

Example 4.6 (the board matrix)**.** For the board matrix A of Definition 2.5,

$$A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}.$$

The entry $(A^2)_{13} = 1$ counts the walk n_1, n_2, n_3. The entry $(A^2)_{22} = 2$ counts the walks n_2, n_1, n_2 and n_2, n_3, n_2. The entry $(A^3)_{12} = 2$ counts n_1, n_2, n_1, n_2 and n_1, n_2, n_3, n_2. The diagonal of A^2 is $1, 2, 1$, the degrees. The entry $(A^2)_{12} = 0$, although n_1 and n_2 are adjacent, because no walk of length exactly 2 joins them: the map has no triangle. Exact length is what matters.

For the drawing of fig. 1(b) read as three roads (convention 3) the matrices are

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}.$$

Here $(A^3)_{11} = 2$ counts the closed walks n_1, n_2, n_3, n_1 and n_1, n_3, n_2, n_1. All these matrices were computed by exact integer arithmetic and the walks were listed by direct enumeration (Appendix B).

4.5 Walks are not paths

In plain words. The table counts trips. A trip may go to a neighbour town and come straight back. Such a trip visits a town twice, so it is not a trip that visits no town twice. The table can therefore show a number where no such trip exists.

Proposition 4.7 (walks, not paths)**.** *The entry $(A^k)_{ij}$ counts walks, and in general neither paths nor trails. In particular, if n_i has a neighbour, then $(A^2)_{ii} = d(n_i) > 0$ although there is no path of length 2 from n_i to n_i.*

Proof. A path between n_i and itself has pairwise distinct vertices with first and last equal, so it has one vertex and length 0 (Definition 2.15). By Proposition 4.3(4), $(A^2)_{ii} = d(n_i)$, which is positive when n_i has a neighbour. The walks it counts are n_i, n_t, n_i, and each of these uses the edge $n_i n_t$ twice, so none is a trail. On the board map, $(A^3)_{12} = 2$ (Example 4.6), but a graph with 3 vertices has no path of length 3 by Proposition 3.4(c); and neither of the two walks counted uses its edges only once. □

Table 3 shows the difference on a larger map: the matrix entry follows the walk column and no other. Paramadevan and Sotheeswaran state in their Theorem 1 that the entry counts "different paths" of the given length [5, p. 16]. With "path" in the course or the English sense

Table 3: On the five-town map (fig. 2), the numbers of walks, trails and paths of length k from n_1 to n_5, and the entry $(A^k)_{15}$. The last column equals the first count in every row.

k	walks	trails	paths	$(A^k)_{15}$
1	1	1	1	1
2	1	1	1	1
3	8	3	3	8
4	16	7	3	16
5	72	6	0	72
6	192	6	0	192
7	704	16	0	704
8	2112	0	0	2112

this is not correct as written, by Proposition 4.7; if their word is read as "walk" it agrees with Theorem 4.4. We use their paper for the statement of the test in Section 5 only.

What can be said about paths is this. If $i \neq j$ and $(A^k)_{ij} \geq 1$, then there is a walk of length k (Theorem 4.4), hence a path between n_i and n_j of length at most k (Lemma 3.2). The value of the entry does not give the number of paths.

5 Question 2: deciding connectivity from A

5.1 Connected graphs and components

In plain words. In some countries you can drive from any town to any other town. In others, some town is cut off from the rest.

Definition 5.1 (connected; textbook Definition 2.1.1)**.** A graph G is *connected* if for any two vertices u and v there is a path between u and v [7, Definition 2.1.1, p. 48]. The board states the same definition: a graph is called connected if for any n_i, n_j there exists a path between the nodes [2].

Remark 5.2. The textbook's word "path" in Definition 2.1.1 is its trail, while the board and the course use the path of Definition 2.15. Corollary 3.3 shows that this does not change which graphs are connected: a walk, a trail and a path between two vertices exist for the same pairs. A graph with one vertex is connected, since there are no two different vertices to join.

Example. The board map of fig. 1(a) is connected: the paths n_1, n_2, n_2, n_3 and n_1, n_2, n_3 join the pairs of vertices, and every vertex is joined to itself by a path of length 0.
Non-example. The map of fig. 1(c) is not connected: n_3 has no neighbour, so no walk and hence no path joins n_3 to n_1 (Corollary 3.3).

In plain words. Towns that can reach each other form an island. Towns on different islands cannot reach each other. A country with one island is one where every town can be reached from every other.

Definition 5.3 (connectivity component; textbook, p. 49)**.** The relation "there is a path between u and v" is an equivalence relation on V (reflexive, symmetric and transitive). Its classes induce subgraphs of G, called the *connectivity components* of G [7, p. 49]. The subgraph induced by a set X of vertices has the vertices in X and the edges of G with both ends in X.

Remark 5.4. A graph is connected if and only if it has exactly one component ([7, Remark 2.1.1, p. 49]; also [1, Definition 2.9.12, p. 42]). Two vertices lie in the same component exactly when a path, and by Corollary 3.3 a walk, joins them.

Example. The map of fig. 1(c) has two components: the one with vertices n_1, n_2 and the road n_1n_2, and the one with the single vertex n_3.
Non-example. $\{n_1, n_3\}$ is not the vertex set of a component of the map of fig. 1(c), since no path joins n_1 and n_3.

5.2 The test

In plain words. To see whether one town can reach another, look at trips of one road, of two roads, and so on. Add up the tables for all these trips. A zero in a place means no such trip joins the two towns, whatever the number of roads. We only need to try a few numbers of roads. A trip that visits no town twice has at most one road less than there are towns.

Theorem 5.5 (connectivity test)**.** *Let G be a simple graph with $N \geq 2$ vertices and adjacency matrix A, and put $S = A + A^2 + \cdots + A^{N-1}$. Then G is connected if and only if $S_{ij} > 0$ for every $i \neq j$.*

The textbook has no matrix test (Section 1.4). Paramadevan and Sotheeswaran state this test as their Theorem 2 [5, p. 17]. The proof below is ours and uses Theorem 4.4, Lemma 3.2 and Proposition 3.4(c).

Proof. Connected implies $S_{ij} > 0$. Let G be connected and let $i \neq j$. By Definition 5.1 there is a path between n_i and n_j (by Corollary 3.3, whichever of the readings of "path" is used); let k be its length. Since $i \neq j$, $k \geq 1$, and by Proposition 3.4(c) $k \leq N - 1$. A path is a walk, so by Theorem 4.4 $(A^k)_{ij} \geq 1$. Every power of A has non-negative entries (Definition 4.1), so $S_{ij} \geq (A^k)_{ij} \geq 1$.

$S_{ij} > 0$ for all $i \neq j$ implies connected. Fix $i \neq j$. The number S_{ij} is the sum of the non-negative integers $(A^k)_{ij}$ for $k = 1, \ldots, N-1$, so one of them is positive: some k with $1 \leq k \leq N-1$ has $(A^k)_{ij} \geq 1$. By Theorem 4.4 there is a walk of length k from n_i to n_j, and by Lemma 3.2 a path between n_i and n_j. A vertex is joined to itself by the path of length 0. So any two vertices are joined by a path, and G is connected. □

Remark 5.6. The step from a positive entry to a path is where care is needed. The entry counts walks (Proposition 4.7), and a walk need not be a path. Lemma 3.2 supplies the step, and Proposition 3.4(c) shows that powers up to $N - 1$ are enough. In our reading, the proof of Theorem 2 of Paramadevan and Sotheeswaran [5, p. 17] treats the entries as counts of paths and passes from a nonzero entry to a path without this step; this is an interpretation of their text, and we use their paper for the statement of the test only. For $N = 1$ the sum S is empty and the single-vertex graph is connected, so the test is stated for $N \geq 2$.

In plain words. There is a neater way to write the same test. Add a stay-at-home option to the road table. Then multiply that table by itself a number of times, one less than the number of towns. The test then reads: no zero anywhere.

Corollary 5.7 (the form with $(I+A)^{N-1}$)**.** *A simple graph G with $N \geq 1$ vertices is connected if and only if the matrix $(I + A)^{N-1}$ has no zero entry.*

The idea is that I and A commute, so $(I + A)^{N-1}$ expands by the binomial theorem into $\sum_{k=0}^{N-1} \binom{N-1}{k} A^k$, a sum with positive coefficients of the same powers as in S plus I. The diagonal is positive because of I, and an off-diagonal entry is positive exactly when the same entry of S is. The full proof is in Appendix A. No source that we read states the test in this form with a proof; the result is derived in this note.

5.3 Worked examples

In plain words. First the board country, with three towns in a row. We list the road table, the table for trips of two roads, and their sum. Then we look for zeros.

Example 5.8 (a connected map)**.** The board map has $N = 3$, so $S = A + A^2$ and the other form is $(I + A)^2$. The matrices are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

$$I + A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

Every off-diagonal entry of S is positive and $(I + A)^2$ has no zero entry, so by Theorem 5.5 and Corollary 5.7 the map is connected, as it should be. The entry $S_{13} = 1$ comes from $(A^2)_{13} = 1$, the walk n_1, n_2, n_3. The entry $(A^2)_{12} = 0$ although n_1 and n_2 are adjacent; the entry $A_{12} = 1$ makes up for it in S. This is why the powers are summed and a single power is not enough. As a check of Corollary 5.7, $(I + A)^2 = I + 2A + A^2$ here.

In plain words. Now a country where one town has no road. We do the same calculation and find zeros.

Example 5.9 (a map that is not connected)**.** Take the map of fig. 1(c), with the single road $n_1 n_2$ and n_3 alone. Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$I + A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The matrix S has zeros at $(1, 3)$, $(2, 3)$ and their mirror images, and $(I + A)^2$ has zeros in the same places. By Theorem 5.5 and Corollary 5.7 the map is not connected. By Corollary 5.11 below the components are $\{n_1, n_2\}$ and $\{n_3\}$.

In plain words. For the drawing on the board, read with a third road, every pair of towns is joined by a road. We expect no zeros.

Example 5.10 (the drawing read as three roads)**.** For the map of fig. 1(b) every entry of S is 2 and every entry of $(I + A)^2$ is 3. The map is connected, which is also visible directly.

5.4 Components, distance and sharpness

In plain words. The table of sums also tells which towns share an island. A zero in a place means the two towns are on different islands.

Corollary 5.11 (components from S)**.** *For $i \neq j$, the vertices n_i and n_j lie in the same component of G if and only if $S_{ij} > 0$. So the pairs with $S_{ij} = 0$ are exactly the pairs in different components.*

Proof. If n_i and n_j are in the same component, a path joins them (Remark 5.4), and the first half of the proof of Theorem 5.5, applied to this pair alone, gives $S_{ij} > 0$. If $S_{ij} > 0$, the second half of that proof gives a path between n_i and n_j, so they are in the same class of the relation of Definition 5.3. □

Remark 5.12 (distance). For distinct vertices in the same component, $d(n_i, n_j)$ is the least $k \geq 1$ with $(A^k)_{ij} > 0$. Indeed, a walk of length k exists exactly when the entry is positive (Theorem 4.4), and the shortest walk has the length of the shortest path (Proposition 3.4(a)). On the board map, $A_{13} = 0$ and $(A^2)_{13} = 1$, so $d(n_1, n_3) = 2$, in agreement with Example 3.6.

Remark 5.13 (the limit $N-1$ cannot be lowered). This is an experimental finding, a computation on four graphs, and it illustrates Proposition 3.4(c) without proving it. On the path graphs with $N = 3, 4, 5, 6$ vertices (one road between consecutive towns), the first power with a positive $(1, N)$ entry is A^{N-1} in each case [4]. A limit below $N - 1$ would miss such a pair.

5.5 A computer check

This is an experimental finding, not a proof. For every labelled simple graph with $N = 1, 2, \ldots, 6$ vertices (1, 2, 8, 64, 1024 and 32768 graphs, of which 1, 1, 4, 38, 728 and 26704 are connected), both matrix tests of this section gave the same answer as a breadth-first search from n_1 [4]. The proofs above do not rest on this check.

5.6 Breadth-first search, for comparison

In plain words. Stand in town 1. Visit all its neighbour towns. Then visit the neighbours of those. Note each town once. If every town was visited, then every town can be reached from town 1.

Breadth-first search starts with the set $\{n_1\}$ and repeatedly adds every n_j with $A_{kj} = 1$ for some n_k already in the set. When nothing can be added, G is connected exactly when the set contains all N vertices. The vertices added are those joined to n_1 by a path, so by Remark 5.4 the set is the component of n_1. Grinberg states that in a breadth-first search tree the tree path from the root is a shortest path [1, p. 176]; using the search to decide connectivity is our reading, not a statement of that source. By our own count, forming S takes $N - 2$ matrix products, whereas the search reads the row of each visited vertex at most once. On the maps of Example 5.8 and Example 5.9 it visits n_1, n_2, n_3 and n_1, n_2 respectively [4].

6 Discussion: limits and open points

What the results do not say. The entry $(A^k)_{ij}$ gives the number of walks, not of paths (Proposition 4.7), and for connectivity only whether it is positive matters. The size of an entry is not a measure of how well two vertices are joined: table 3 shows entries growing to 2112 at $k = 8$ while no path on this five-town map has more than 4 edges (Proposition 3.4(c)). The test of Theorem 5.5 decides connectivity; it does not list the paths.

Scope of the proofs. Both main results are stated and proved for finite simple undirected graphs (convention 1). The proof of Theorem 4.4 identifies a walk with its vertex sequence and an edge with the entry 1, and we do not claim the result for graphs with loops or multiple edges. For directed graphs the matrix is not symmetric and "connected" splits into more than one notion, so the test is not claimed there; Grinberg's version of the walk-counting theorem covers directed graphs [1, Theorem 4.5.10].

Conventions that rest on us. Four readings are assumptions or choices rather than facts read from a source: that the mark above A on the board is a power k (convention 2); that the matrix and not the drawing is the example (convention 3); that the course word "path" means no repeated vertex (convention 5), which is the user's statement and agrees with the textbook's "simple path"; and that a walk of length 0 is allowed (convention 6). Corollary 3.3 shows that the second reading of "path" does not change which graphs are connected, so the test is the

same under any of the three. Nothing in this note claims what the lecturer intended beyond what the board photographs and the user's text say.

Where sources disagree. The textbook and English usage give the word "path" different meanings (table 2). Paramadevan and Sotheeswaran's Theorem 1 speaks of "paths" where the entry counts walks (Section 4), and the proof of their Theorem 2 omits the step from a walk to a path (Remark 5.6). We resolved both by proving the statements here.

Strength of the evidence. The absence of the walk-counting theorem and of the matrix test from the textbook rests on a search of its text layer (Section 1.4). The computer checks (the exhaustive comparison up to $N = 6$, the four path graphs of Remark 5.13) cover finitely many cases and support the proofs without replacing them. Grinberg's text is a long-lived lecture-notes preprint, and the encyclopedia page is used only for the English names of the walk types.

7 Conclusion

Q1. For a finite simple undirected graph with adjacency matrix A, the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j (Theorem 4.4), for every $k \geq 0$. Walks may repeat vertices and edges, so the entry does not count paths or trails (Proposition 4.7). For example, on the board map $(A^2)_{11} = 1$ counts the walk n_1, n_2, n_1. This is the textbook's walk, Definition 1.2.10; the textbook has no such theorem, so the proof is given here.

Q2. A graph with $N \geq 2$ vertices is connected if and only if every off-diagonal entry of $S = A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 5.5), and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Corollary 5.7). The proof uses the walk-counting theorem, the lemma that a walk contains a path of no greater length (Lemma 3.2), and the fact that a path has at most $N - 1$ edges (Proposition 3.4(c)). On the board map every off-diagonal entry of S is 1, so the map is connected; on the map with an isolated town S has zeros, so it is not (Example 5.8, Example 5.9).

Q3. A path between n_i and n_j is a walk $w_0, \ldots, w_k$ from n_i to n_j whose vertices are pairwise distinct (Definition 2.15). This is the textbook's simple path, Definition 1.2.13. The textbook's own word "path", Definition 1.2.12, means a walk with no repeated edge, which this note calls a trail. By Corollary 3.3 the choice among walk, trail and path does not change which graphs are connected.

References

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[2] *Lecture whiteboard, photo 1.* Local file sources/whiteboard-1.jpg. Photo supplied by the user.

[3] *Lecture whiteboard, photo 2.* Local file sources/whiteboard-2.jpg. Photo supplied by the user; same board as photo 1 from another angle.

[4] *m3_connectivity.py: matrices, tests and exhaustive check.* Local file scripts/m3_connectivity.py, output in scripts/m3_connectivity.out. Written for this paper.

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A Derivation of the form with $(I + A)^{N-1}$

This appendix proves Corollary 5.7: a simple graph G with $N \geq 1$ vertices is connected if and only if $(I + A)^{N-1}$ has no zero entry. It uses Theorem 5.5 and the notation of Definition 4.1.

Proof of Corollary 5.7. The identity matrix commutes with every matrix, so I and A commute and the binomial theorem applies:

$$(I + A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \tag{2}$$

where $\binom{N-1}{k}$ is the number of ways to choose k items from $N - 1$ items, and is at least 1 for $0 \leq k \leq N - 1$. By Definition 4.1 every A^k has non-negative integer entries, so every term of (2) is non-negative.

Diagonal entries. The term with $k = 0$ is $\binom{N-1}{0} I = I$, which contributes 1 to every diagonal entry. All other terms are non-negative, so every diagonal entry of $(I + A)^{N-1}$ is at least 1 for every graph.

Off-diagonal entries. For $i \neq j$ the term $k = 0$ contributes 0, so

$$\big((I + A)^{N-1}\big)_{ij} = \sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}.$$

The coefficients are positive and the terms $(A^k)_{ij}$ are non-negative, so this sum is positive exactly when at least one $(A^k)_{ij}$ with $1 \leq k \leq N - 1$ is positive. That is exactly when $S_{ij} > 0$, because S_{ij} is the sum of the same non-negative terms.

Conclusion. Hence $(I + A)^{N-1}$ has no zero entry if and only if $S_{ij} > 0$ for all $i \neq j$. For $N \geq 2$ this is connectivity by Theorem 5.5. For $N = 1$ the matrix is $I = (1)$, which has no zero entry, and a graph with one vertex is connected (Definition 5.1). □

On the board map, (2) reads $(I + A)^2 = I + 2A + A^2$, which was checked by exact integer arithmetic (Example 5.8). No source that we read states or proves the test in this form, so this derivation is the only support for it.

B Sources, checks and how to rerun the computations

Sources and what was read. The *course textbook* [7] is a local file supplied by the user. Its title is in Armenian script; the reference list gives a transliteration because the document is typeset with a Latin font. We viewed the page images of printed pages 11, 21, 22, 23, 48 and 49 before quoting definitions, and read the text layer of pages 16 and 22–23. The rest of the book was searched, not read. The two *board photographs* [2, 3] show the same board from two angles and were read together. *Grinberg's lecture notes* [1] were read in Section 2.9, in Theorem 4.5.10 and its proof, and in the passage on breadth-first search. *Paramadevan and Sotheeswaran* [5] were read in Section 3.1 (Theorems 1 to 3 and their proofs). The *encyclopedia page* [6] was read as raw wikitext on 2026-11-05 and is used only for names. The *computer check* [4] is a short program written for this note.

What is ours. The statements and proofs of Lemma 3.2, Corollary 3.3, Proposition 3.4, Proposition 4.3(2)–(4), Theorem 4.4, Theorem 5.5, Corollary 5.7, Corollary 5.11 and Remark 5.12 are proved in this note. Where a source states the same result, the text says which one. The reading of the board is recorded as conventions 1 to 3 in Section 1.3.

Rerunning the numbers. Every matrix, count and example in the text is recomputed, with exact integers, by the program `appendices/computations.py` in the paper folder. From that folder, run `python3 -I appendices/computations.py`; it takes about twenty seconds, and its output is saved as `appendices/computations.out.txt`. The program lists the matrices of Example 4.6, Example 5.8, Example 5.9 and Example 5.10, the walks that these examples name, the statuses (walk, trail, path, closed) of the examples in Section 2, the counts of table 3, a check of the binomial identity, a consistency check of the cutting rule of Lemma 3.2 on all short walks of the five-town map, and the exhaustive comparison with breadth-first search for $N \leq 6$. These checks support the proofs and do not replace them.